

Numerical Application of Quartapelle's Projection Condition (Time Invariant Case)

Tags: #Numerical-Method

Note

Quartapelle's projection condition for the *Stream function-Vorticity formulation* shows that the distribution of vorticity ω is controlled by the boundary conditions of the stream function ψ , and it is a integral constraint. Thus, he provided some numerical schemes to solve the problem.

Related note: [[Quartapelle's Projection Condition]]

The Numerical Scheme

The goal of computation is to find the vorticity field satisfying the projection condition, which is

$$\int_V \zeta \eta_\alpha dV = \oint_{\partial V} (b\eta_\alpha - a \frac{\partial \eta_\alpha}{\partial n}) ds, \quad \nabla^2 \eta_\alpha = 0$$

However, the theorem requires the information of **all** harmonic functions in the domain. To put this into practice, we will need to limit the number to functions with the boundary points number, thus $\alpha = 1, 2, \dots, N$, where N is the number of boundary points (Actually they are linear spaces with dimension N equal to the number of grid points at the boundary.) . The harmonic functions η_α are linearly independent, thus they belong to the basis of discrete harmonic space. Those functions can be set as

$$\nabla^2 \eta_\alpha = 0, \quad \eta_\alpha|_S = \delta_{\beta\alpha}$$

where $\delta_{\beta\alpha}$ is Kronecker symbol. In practice, this means that each boundary point corresponds to a (discrete) harmonic function, and it only has the value 1 at that point. The value of that function on other boundary points vanishes. The harmonic functions do not reconstruct the whole vorticity field; they span the harmonic correction needed to enforce the projection of the vorticity onto the discrete harmonic space.

When ζ^m and ψ^m are known, and we assume that $(\zeta^{m+1}, \psi^{m+1})$ is the solution of the problem,

such that

$$\nabla^2 \zeta^{m+1} = \text{Re } J(\zeta^m, \psi^m), \quad J(\zeta, \psi) = \mathbf{u} \cdot \nabla \zeta \quad (1)$$

$$\int \zeta^{m+1} \eta_\alpha dV = \oint (b\eta_\alpha - a \frac{\partial \eta_\alpha}{\partial n}) ds \quad (2)$$

$$\nabla^2 \psi^{m+1} = \zeta^{m+1} \quad (3)$$

$$\psi^{m+1}|_S = a \quad (4)$$

The procedure of resolution can be described as: - To solve the problem, we start with imposing ζ^0 and ψ^0 in the equation (1), then solve it to get ζ_1 . - With the value ζ_1 , we can set the update scheme

$$\zeta^{m+1} = \zeta_1 + p_\beta^{m+1} \eta_\beta$$

where $p_\beta^{m+1} \eta_\beta = \sum_\beta^N p_\beta^{m+1} \eta_\beta$. - If we substitute the update scheme into the equation (2), then

$$\int (\zeta_1 + p_\beta^{m+1} \eta_\beta) \eta_\alpha dV = \oint (b\eta_\alpha - a \frac{\partial \eta_\alpha}{\partial n}) ds$$

Thus

$$\int p_\beta^{m+1} \eta_\beta \eta_\alpha dV = - \int \zeta_1 \eta_\alpha dV + \oint (b\eta_\alpha - a \frac{\partial \eta_\alpha}{\partial n}) ds$$

If we take all boundary points into account, then this expression can be reformulated as a linear system such that

$$\mathbf{A} \mathbf{p}^{m+1} = -\mathbf{c}^{m+1}$$

where

$$A_{\alpha\beta} = \int \eta_\alpha \eta_\beta dV,$$

$$\mathbf{p}^{m+1} = (p_1^{m+1}, \dots, p_N^{m+1}),$$

$$-c_\alpha^{m+1} = - \int \zeta_1 \eta_\alpha dV + \oint (b\eta_\alpha - a \frac{\partial \eta_\alpha}{\partial n}) ds$$

We just need to solve this linear system to get the values of p_β^{m+1} and repeat this procedure. $\mathbf{c}^{m+1}$ must be computed in each iteration, but $A_{\alpha\beta}$ can be computed once. - We can also set some numerical residuals to observe the convergence in the iterations.

Thought

- The condition

$$\eta_\alpha|_S = \delta_{\beta\alpha}$$

should be interpreted as a Kronecker basis condition in the discrete boundary space, not as a continuous Dirac distribution. It is a discrete analogue of a point-supported boundary basis, whose harmonic extension forms a basis of the discrete harmonic space.

- Theoretically, we will need to test all harmonic functions in the domain to obtain the real distribution of the vorticity field with this method, but in practice it is not feasible. Thus

the number of harmonic functions is restricted by the number of boundary cells, which also means that the accuracy of the scheme highly depends on the number of boundary cells.

Reference:

- L. Quartapelle and F. Valz-Gris, *Projection Conditions on the Vorticity in Viscous Incompressible Flows*, International Journal For Numerical Methods in Fluids, Vol.1, 129-144(1981)